

EAS-VGGT: Supplementary Material

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1. Data and I/O Contract

Every Actor artifact separates build inputs from target arrays and stores scene, log, track, pose, and measurement provenance. The route role contains three logs, 31 scenes, 310 Actors, 816,442 build points, and 284,030 target rays. The source role contains three logs and 17 scenes. Selective extraction from official nuScenes shards 02 and 10 recovers 677/677 required source keyframes (470,123,520 bytes). Source ActorBundleV2 and EAS corpora independently materialize the same 80 Actors and share fingerprint `2ec626e1...fcc4c1c`; 50 are hazardous, with 265,177 build points and 94,615 query rays. RGB recovery extracts only predeclared JPEG paths and records archive and image SHA-256 values.

2. Canonical Runs

The primary development adapter is `20260907T183000Z__e2-canonical-late-fusion-s7202-r1`. The route reliability run is `20260907T223000Z__e5-route-calibrated-s7501-r2`. It selects $\alpha = .35$ by Brier and then NLL on a grid from zero to one; the source configuration stores this value, checkpoint hashes, code hashes, scene/sample identities, and the route-summary hash before source payload access.

The ordered-return run is `20260907T195500Z__e2-ordered-return-s7203-r2`. Its held-out set has 16,384 rays; 1,734 contain an undetected front blocker. The separated model obtains no-return MAE .00131 on that subset versus .37441 for the single-hazard reader. After the order intervention, NLL remains .67563 versus .93539.

The appearance bridge is `20260907T213500Z__e3-render-bridge-s7302-r2`. All six held-out views improve Actor PSNR. Mean change outside the Actor support is .000707 versus .02836 inside, while physical and trajectory parameters are not trainable from RGB. The SE(3) run is `20260907T191500Z__e4-se3-rigid-trajectory-s7401-r4`.

3. Foundation Model Provenance

VGGT uses the official `facebook/VGGT-1B` checkpoint; π^3 uses the released `yyfz233/Pi3X` checkpoint. MapAnything is pinned to repository commit `3d10cf7a...` and checkpoint SHA-256 `981f060c...ed5a3a0`. It receives RGB, calibrated intrinsics, and metric OpenCV camera-to-world poses through its native preprocessor. The diagnostic uses BF16 memory-efficient inference and peaks at 7.15 GiB allocated GPU memory.

4. Frozen Source Outcome

The source cache processes the exact 12 locked sample tokens and peaks at 7.61 GiB. It finds eight Actor-pose matches but no candidate with a camera observation. The formal evaluator reaches all four arms, whose evidence tensors are empty, then correctly refuses to serialize non-finite means. The failed attempt is retained. A separate target-free support audit verifies the original lock and writes the canonical verdict `frozen_confirmation_inconclusive_insufficient_visual_support`; it loads no checkpoint or label and performs no selection. A future evaluation must freeze a minimum camera/Actor support rule before opening source quality.

5. Failure Boundaries

Engineering recoveries include official RGB and LiDAR selective I/O, dtype conversion for MapAnything camera prompts, and strict support reporting for empty cohorts. Scientific negative results remain visible: early visual fusion overfits, π^3 visual evidence hurts on development, raw VGGT fusion harms route Brier, and the frozen source set is not evaluable. None is converted into a positive claim by changing scenes or thresholds.